

Core Java E-Commerce Project with source code and readme file

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Objective: Develop a console-based E-Commerce system where users can browse products, add to cart, checkout, and process orders, with data stored in a MySQL database using JDBC.

Java's Key Features & Concepts

1. Object-Oriented Programming (OOP)

1. Classes: **Product, User, Cart, Order, OrderItem, OrderService.**
2. Implements **Encapsulation, Inheritance, Polymorphism, and Abstraction.**

2. String Manipulation

1. Enable product search by name.
2. Validate email formats during user registration.

3. Exception Handling

1. Handle errors like invalid product selection, insufficient stock, and payment failures.
2. Use custom exceptions to improve error messages.

4. Multi-threading (ExecutorService for Order Processing)

1. Use ExecutorService (ThreadPool) for parallel order processing.
2. A fixed number of worker threads will handle orders efficiently.

5. Collections Framework

1. Use List to store products and orders.
2. Use Map for the shopping cart (Product → Quantity).

6. Java 8 Features

1. Streams & Lambda Expressions for filtering and sorting products.
2. Optional for handling null values in user data retrieval.

7. JDBC (Database Operations)

1. CRUD operations for Products, Users, and Orders using MySQL & JDBC.
2. Implement transactions for atomicity during order placement.

8. Order Workflow

1. User Registration/Login
2. Product Browsing & Filtering
3. Add to Cart & Checkout
4. Order Placement & Stock Update
5. Order Processing Using Executor Service (Thread Pool)
6. Order Shipment Simulation

Updated Order Processing Approach: Executor Service

7. Use Executor Service with a Fixed Thread Pool for handling orders.
8. When an order is placed, it is submitted to a worker thread for processing.
9. Each thread handles an order validation, stock update, and payment processing.
10. Once completed, the thread is released back to the pool.

Functional Flow

1. User

Registration & Login

User Registration:

1. User enters name, email, password → Validate & store in Database (JDBC).
2. If email already exists, show error.

User Login:

3. Validate credentials → Fetch user details from Database.
4. If invalid → Show "Invalid Credentials" error.

2. Product

Browsing & Search

View All Products:

1. Fetch available products from Database (JDBC SELECT Query).

Search & Filter:

2. Use Java 8 Streams to filter based on name, category, price.

Check Availability:

3. Show "In Stock" or "Out of Stock".

3. Add to Cart &

Manage Cart Add

Product to Cart:

1. Store selected items in Map<Product, Quantity>.
2. Validate stock before adding.

View Cart:

3. Display items, price, total cost.

Remove Item:

4. Remove specific product from Map.

Modify Quantity:

5. Adjust quantity and update total price.

4. Checkout & Order Placement

Validate Cart:

1. Ensure products are in stock before proceeding.

Confirm Payment:

2. Simulate payment (Assume success for now).

Store Order in Database:

3. Insert order details into Orders Table (JDBC Transaction).

5. Order Processing (Multi-threading

withExecutorService) Submit Order for Processing:

1. Use ExecutorService (ThreadPool) to handle multiple orders concurrently.

Deduct Stock & Update Order Status:

2. Reduce stock count in Database.

Simulate Order Packing & Shipment:

3. Threads simulate different stages (Packing → Shipping).

Handle Order Failures:

1. If stock is insufficient, **rollback transaction**.

2. Order Completion

& Shipment Mark Order

as "Shipped"

1. Update order status in DB.

Display Confirmation to User

2. Show estimated delivery time.

End Order Processing Thread

3. ExecutorService releases thread after task completion.

Thank you!

This is a complete project what you
need to do in this project

THANKYOU

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Next page how can you Build this whole
project [readme]

Java E-Commerce Console Application

A complete console-based E-Commerce system developed using Core Java, JDBC, and MySQL. This project simulates the key features of an online shopping platform.

✓ Features

- User Registration & Login (with email uniqueness check)
- Product Browsing and Searching (Java 8 Streams)
- Shopping Cart (Add, View, Remove Items)
- Checkout & Order Placement (Multi-threading using Executor Service)
- Stock updates in database after checkout
- Order Processing with Shipment simulation
- Order History (View past orders with total cost)

💻 Technologies Used

- Java 8
- JDBC (Java Database Connectivity)
- MySQL Database
- Maven (Dependency Management)
- Console-based User Interface

📁 Folder Structure

```
src/
├── main/
│   ├── java/
│   │   ├── model/ # POJO classes: User, Product, CartItem, Order
│   │   ├── service/ # Business logic: UserService, ProductService, CartService,
│   │   │   OrderService
│   │   ├── db/ # DB connection utility
│   │   └── main/ # Entry points: Main.java, UserMain.java, ProductMain.java,
│   │       CartMain.java, CheckoutMain.java
│   └── pom.xml # Maven configuration
```

🔧 Setup Instructions

1. Install MySQL and create the database `ecommerce\_db`
2. Run the SQL schema to create tables: users, products, orders, order_items
3. Insert some sample product data
4. Clone or download this project and open it in Eclipse (as a Maven project)
5. Update `DBConnection.java` with your DB credentials
6. Run `Main.java` to start the full flow

Database Tables

- `users` — stores registered users
- `products` — stores product catalog
- `orders` — stores placed orders
- `order\_items` — stores product line-items for each order

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If you want to source code and full structure, go to git hub in project repo and check it

Congratulation project is completed

THANKYOU

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